Student Name: Student ID:

Question 1 (30 Marks)

Answer the following questions about Python Programming.

1.1 What are the differences between lists and sets in Python? (5 Marks)

Ans:

List: A list contains an arbitrary number of elements that are stored in sequential order. The elements don't need to be of the same type. The list is mutable.

Sets: The set represents a collection of unique elements. The set can be created by listing its elements between curly brackets. Sets are mutable.

Type	Example	Description	Can we change the content within this collection?	Are the items in the collection ordered stored?
list	[1, 2, 3]	List: an ***mutable***, ***ordered*** collection	Yes – mutable	Yes – ordered
tuple	(1, 2, 3)	Tuple: an ***immutable, ordered*** collection	No – immutable	Yes – ordered
set	{1, 2, 3}	Set: ***Unordered*** collection of unique values	Yes – mutable	No – unordered
dict	{'a':1, 'b':2, 'c':3}	Dictionary: ***Unordered*** (key, value) mapping	Yes – mutable	No – unordered

1.2 What will be the output of the following Python code? (5 Marks)

```
a = [1, 6, 3, 9, 7, 8]
a.sort(reverse=True)
print(a[5:2:-2])
```

Ans: [1, 6]

1.3 What will be the output of the following Python code? (5 Marks)

```
In [3]: a = numpy.arange(16).reshape(4,4) # Define 4 *4 matrix using NumPy library
row = numpy.array([0,1])
col = numpy.array([2,3])
print(a[row,col])
```

Ans: [2, 7]

1.4 What will be the output of the following Python code? (5 Marks)

```
import numpy as np
a=np.arange(3)+4
print(a)
```

Ans: [4 5 6]

1.5 What will be the output of the following Python code? (5 Marks)

```
import numpy as np
a=np.array([[5,6,8,6],
            [1,6,4,8],
            [1,8,5,1]])
print(a.sum(axis=0))
```

Ans: [7 20 17 15]

1.6 What will be the output of the following Python code? (5 Marks)

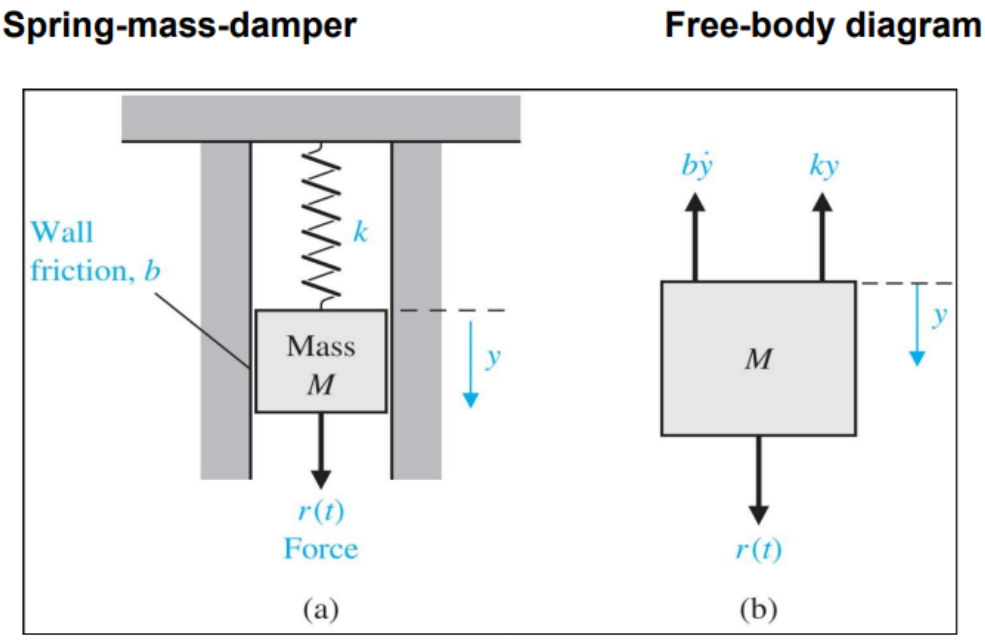
```
def my_print(person = "Joe"):
    message = "Hello, " + person
    print(message)

my_print("Peter")
```

Ans: Hello, Peter

Question 2 (20 Marks)

Write the differential equation of the following mechanical system.



r =Applied force (N), M =mass (kg), y =displacement (m),

b = friction coefficient, k =stiffness of spring (N/m).

Answer:

Newton seconds law of motion $F_M(t) = M\frac{d^2y(t)}{dt^2}$

Elastic Force $F_k(t) = ky(t)$

Wall friction Force $F_f(t) = b\frac{dy(t)}{dt}$

Differential Equation of the system $\sum F(t) = 0$

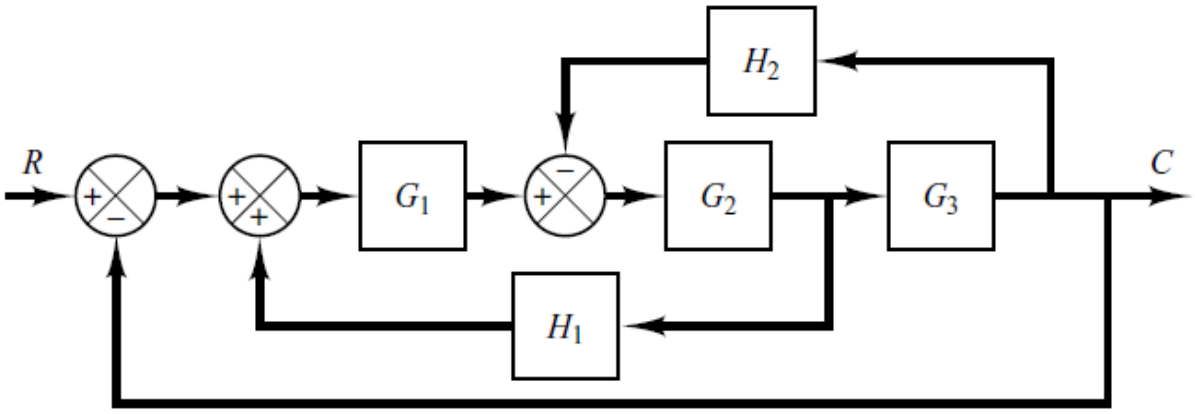
$$r(t) - F_M(t) - F_k(t) - F_f(t) = 0$$

$$r(t) = M\frac{d^2y(t)}{dt^2} + b\frac{dy(t)}{dt} + ky(t)$$

$$r(t) = My''(t) + by'(t) + ky(t)$$

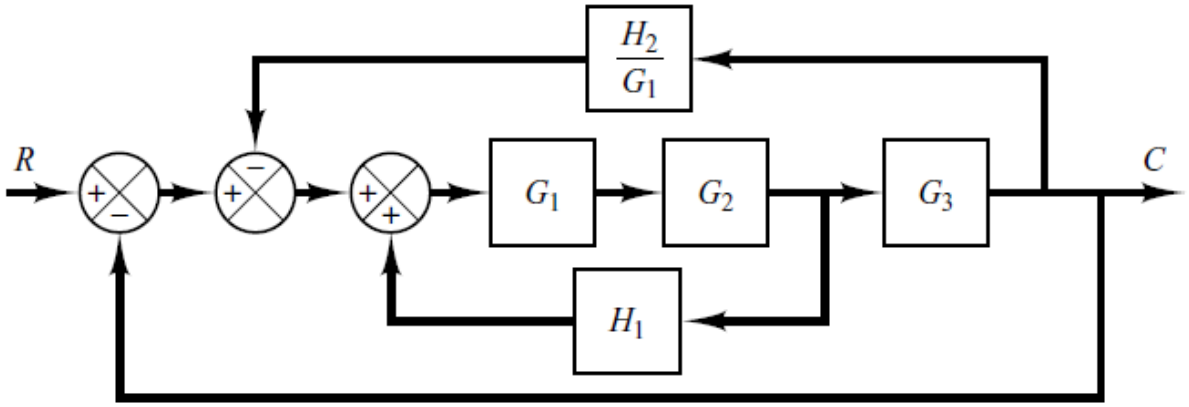
Question 3 (30 Marks)

Reduce the block diagram into simple form and write the final transform function.

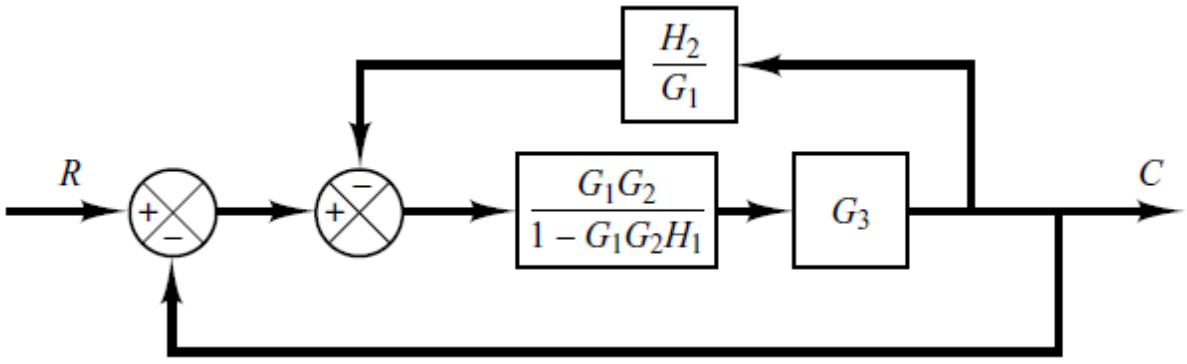


Ans:

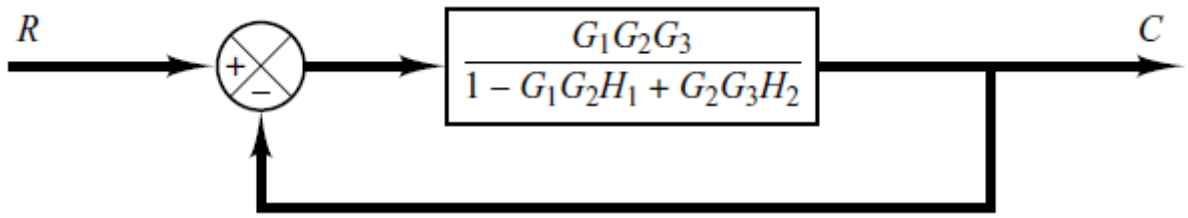
Step-1: By moving the summing point of the negative feedback loop containing H_2 outside the positive feedback loop containing H_1, we obtain the following figure



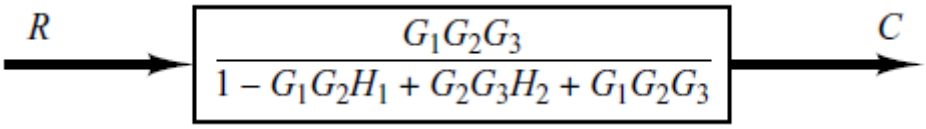
Step-2: Eliminating the positive feedback loop we have



Step-3: The elimination of the loop containing $\frac{H_2}{G_1}$ gives

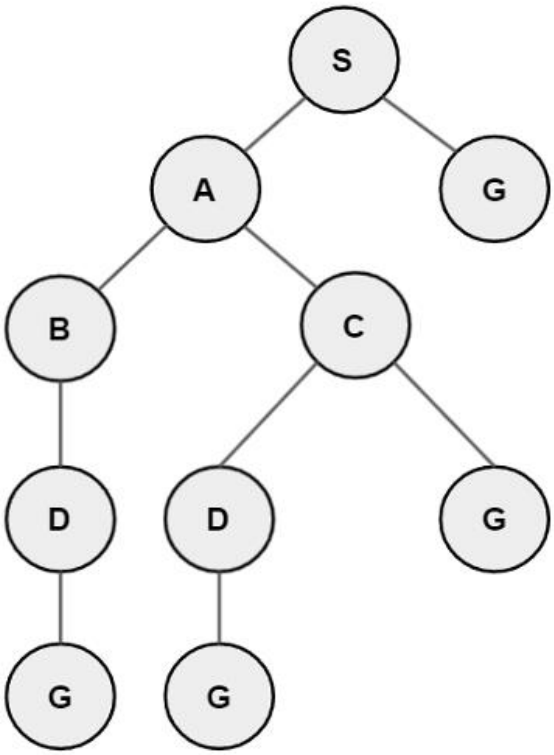


Step-4: Finally, eliminating the feedback loop results in



Question 4 (20 Marks)

Look at the graph below and traverse nodes in the tree starting from node "S" using: 1) **breadth-first search**; 2) **depth-first search**.



ANSWER

1) breadth-first search: S,A,G, B,C,D,D,G,G,G
2) depth-first search: S,A,B,D,G, C,D,G,G,G

Alternative answers are also acceptable:
1) breadth-first search: S,A,G
2) depth-first search: S,A,B,D,G

-END OF QUIZ-